Semantic and Geometric Information Embedding from an IFC Model

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https://github.com/olgapoletkina/ICD_CA_assignmnet/blob/master/olga_poletkina_assignment.ipynb

BIM models contain rich semantic and geometric information that remains largely unexploited by AI methods. This report proposes a framework for transforming IFC data into a graph representation suitable for GNN-based architectural co-pilot systems.

1. Semantic Representation

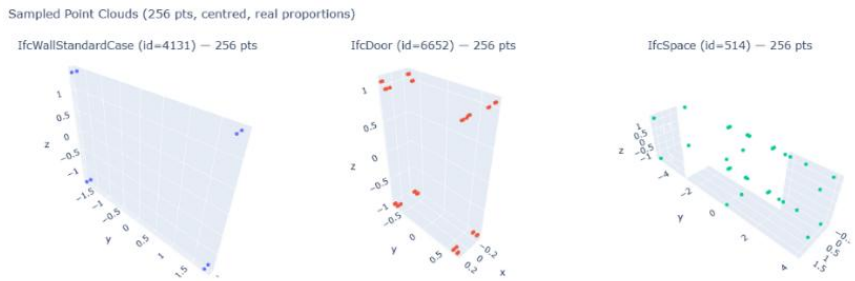
The most meaningful semantic information falls into *several layers*. **Element taxonomy** – the IFC class (IfcWall, IfcDoor, IfcSpace, etc.) is encoded as a one-hot node feature defining each element's role in the building. **Property Sets** – attributes such as FireRating, IsExternal, and LoadBearing are pivoted into a per-node feature vector with missing values imputed by any relevant approach (median, mean, etc.). **Relational semantics** – IFC relationship objects become typed directed edges that encode design intent: a door is linked to a wall because it is hosted by it, and a room (IfcSpace) is connected to walls because it is surrounded by them in the model. The final node feature vector concatenates the IFC type, selected Pset attributes, and optionally *direct attributes of the IFC object*, for example, the element name, which can be represented as a SentenceTransformer embedding with reduced dimensionality. *[1, 2, 3]*

2. Geometric Representation

IFC parametric geometry is tessellated into triangle meshes via `ifcopenshell.geom`. Two complementary encodings are derived:

- **OBB (*n*-dimensional):** each element's mesh is enclosed by an Oriented Bounding Box aligned to the element's own principal axes rather than the global coordinate system. From this box, compact geometric features are extracted: position, size, orientation, etc. Additional features can also be derived, such as area, volume, and aspect ratios – all capturing the element's spatial properties in a form directly useful for machine learning.
- **Shape embedding (*m*-dimensional):** *N* points sampled from the mesh are summarised as a set of order-invariant descriptors (PCA eigenvalue ratios, distance quartiles, vertical spread), then reduced to *m* dimensions by PCA or any relevant dimensionality reduction method.

The combined node vector is *n*+*m*-dimensional, extended further with *Pset* and *name embedding features*.

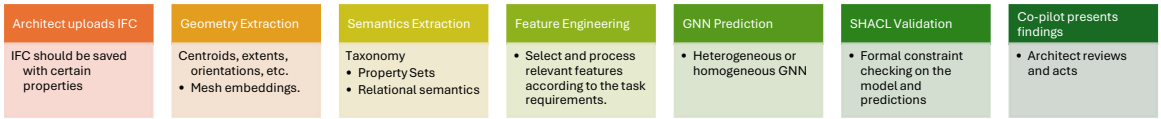


Point clouds by types (samples)

3. Intended AI Capability

The representation enables a range of GNN-based capabilities for an architectural co-pilot: node-level property prediction (fire rating, load-bearing status), edge-level reasoning (spatial adjacency validation), and graph-level tasks (layout quality scoring, design variant generation). All benefit from the relational structure – element requirements depend on their context within the building, which message-passing GNNs capture naturally through typed edges.

The choice of GNN architecture depends on the specific task, but the heterogeneous, multi-relational nature of the BIM graph favours models that explicitly account for edge types - such as R-GCN (separate weight matrices per relation), HGT (Heterogeneous Graph Transformer with type-aware attention), or HAN (Heterogeneous Attention Network). Simpler homogeneous GNNs (GCN, GAT) can also be applied if IFC type and relation type are encoded as node and edge features respectively. Training uses Leave-One-Building-Out cross-validation on several buildings; class imbalance is addressed by weighted cross-entropy and GraphSMOTE. *[4, 5, 6]*

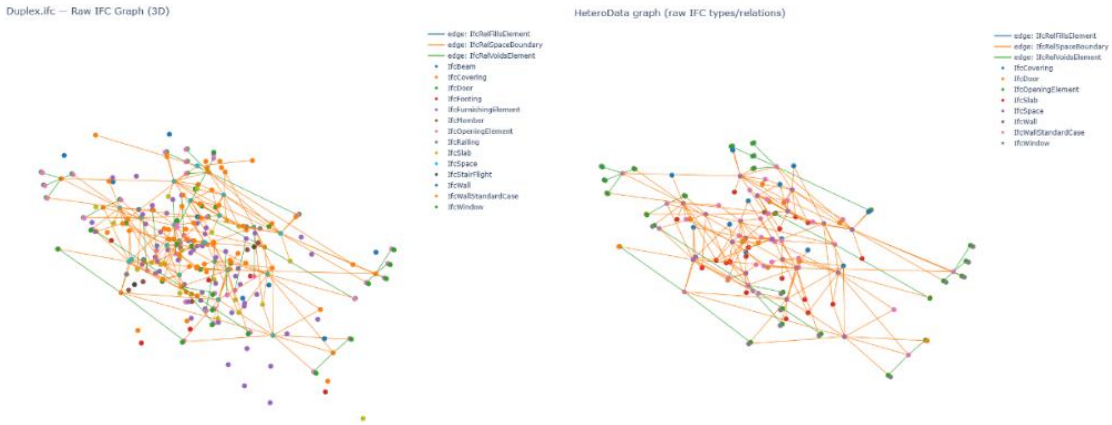


Intended AI Capability schema

4. Data Transformation

Four stages convert a raw IFC file into a PyTorch Geometric object:

- **Geometry extraction:** OBB and point cloud computed per element; geometry-less elements zero-filled if topologically referenced, otherwise excluded.
- **Semantic extraction:** Psets and QTOs parsed and pivoted onto the node table; only properties above a minimum fill rate threshold are retained as training features.
- **Graph construction:** IFC relation types parsed into a typed directed edge list; duplicates and orphan endpoints removed.
- **Cleaning and normalisation:** self-loops removed, largest connected component retained; geometric features normalised to ensure consistent scale across element types.



Graphs extracted and cleaned

5. Symbolic Constraints

Symbolic constraints are integrated at two stages. **Pre-training:** the IFC graph is projected to RDF and validated with SHACL shapes – every *IfcDoor* must fill exactly one opening, every *IfcSpace* must have at least one boundary. Violations flag corrupted topology before the model sees it. **Post-prediction:** GNN outputs are written back to RDF and re-validated; predictions that break a shape are flagged with the exact violated rule, giving the architect a transparent explanation alongside the data-driven suggestion. *As a future direction, SHACL constraints can be encoded as differentiable penalty terms in the loss function, directly guiding the gradient signal. SHACL is preferred over hardcoded rules as it is declarative, W3C-standard, and modifiable by domain experts without touching model code. [8, 9, 10]*

6. Additional Considerations

- **Scalability:** one node per IFC element keeps the graph compact, allowing large multi-building corpora to fit within single-GPU memory.
- **Class imbalance:** GraphSMOTE oversampling and weighted cross-entropy ensure the model detects rare but safety-critical fire-rated elements - false negatives are more dangerous than false positives. *[6]*
- **Generalisation:** per-building z-score normalisation combined with LOBO evaluation prevents scale artefacts and tests true cross-building transfer.
- **Graph representation choice:** HeteroData makes IFC semantics explicit and suits analysis; however, a homogeneous R-GCN also performs well on BIM datasets, as all nodes share one embedding space. *[11]*
- **Workflow integration:** predictions map back to IFC element IDs for direct visualisation in Revit; SHACL reports export as JSON for compliance dashboards.
- **Extensibility:** the pipeline is schema-agnostic – new element types or regulation domains require only updated feature selection and SHACL shapes.

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